

H5. Which of the following sets are equal?

$$A = \{0, 1, 2\}$$

$$B = \{x \in \mathbf{R} \mid -1 \leq x < 3\}$$

$$C = \{x \in \mathbf{R} \mid -1 < x < 3\}$$

$$D = \{x \in \mathbf{Z} \mid -1 < x < 3\}$$

$$E = \{x \in \mathbf{Z}^+ \mid -1 < x < 3\}$$

First, we assess each set and their possible elements.

$$A = \{0, 1, 2\}$$

x is a real number of the set B , such that x is greater than or equal to -1 and less than 3 .

$$A \neq B$$

x is a real number of the set C , such that x is greater than -1 and less than 3 .

$$A \neq C, B \neq C$$

x is an integer of the set D , such that x is greater than -1 and less than 3 .

$$D = \{0, 1, 2\}$$

$$A = D$$

$$B \neq D, C \neq D$$

x is a positive integer of the set E , such that x is greater than -1 and less than 3 .

$$E = \{1, 2\}$$

$$A \neq E, B \neq E, C \neq E, D \neq E$$

Solution: $A = D$